A Weighted Approach to Student Persona Modeling in AI-Driven Educational Simulations

Liyan Altahrawi*, Mouhitha Arella†, Selah Frederickson‡, Lahira Hajdarovic§, Erjon Kadriu¶
School of Innovation, Design, and Engineering
Mälardalen University, Västerås, Sweden
Email: *lai26004, †maa26001, ‡sfn26003, §lhc26001, ¶eku21001@student.mdu.se

***Abstract*—AI-driven classroom simulations can provide controlled environments for teacher training, but believable student behavior requires more than fluent language generation. This study presents a weighted student persona model for a multi-agent classroom simulation, where student agents are represented through knowledge, personality, and affective state variables. The system combines structured student profiles, task-aware interaction logic, heuristic behavioral scoring, and LLM-based response generation to simulate classroom interaction. The main research question investigates to what extent weighted knowledge and personality variables can produce believable and distinguishable student personas in an AI classroom simulation. An evaluation was conducted in which participants read classroom interaction logs and interacted directly with the system, then judged student agents' knowledge, personality traits, affective states, and overall realism. Results suggest that personality-related traits were more consistently recognized than knowledge, misconceptions, or affective states. Evaluators generally perceived student agents as realistic and potentially useful for teacher preparation, although overly mature responses and repetition reduced believability in some cases. The findings indicate that structured student profiles can support believable AI-driven classroom simulations, while misconception modeling, affective expression, and response diversity remain important areas for future improvement.**

***Index Terms*—artificial intelligence, classroom simulation, large language models, multi-agent systems, persona modeling, teacher training**

I. Introduction

Educational systems throughout most of Europe and the United States are facing a persistent shortage of qualified teachers, with projections in some regions, such as Sweden, indicating that issues may accelerate in coming decades [1]. Simultaneously, teacher education programs are often criticized for inadequately preparing student teachers for the complexities of real classroom environments. Student teachers frequently perceive a disconnect between theoretical knowledge and classroom practice, leaving them underprepared for managing student behavior, diverse learning needs, and dynamic teaching situations [2].

On-campus microteaching, in which "preservice teachers teach microlessons to their peers in a simulated environment," provides a valuable learning experience for preservice teachers [3]. However, traditional microteaching is limited due to issues such as anxiety and power imbalances in peer-to-peer role-play scenarios [4]. This highlights the need for scalable and flexible approaches that can effectively supplement traditional practicum models.

Recent advances in artificial intelligence (AI), particularly in large language models (LLMs), offer opportunities to address this gap. Virtual students powered by AI can simulate classroom interactions in controlled and replicable environments, enabling preservice teachers to practice instructional strategies and classroom management [5].

A critical factor in the effectiveness of such systems is emotional realism, as classroom interactions are shaped not only by cognitive processes but also by students' affective states, such as frustration, curiosity, confusion, and motivation, which significantly influence engagement and learning outcomes [6], [7], [8]. However, the representation of these emotional dynamics in AI-based simulations remains underexplored.

The concept of affective generation originates from the field of affective computing, introduced by Picard [9], and refers to a system's ability to initiate or influence emotionally expressive behavior. Incorporating such mechanisms into educational systems may enhance the authenticity and pedagogical value of AI-driven interactions. However, effectively modeling these dynamics in a structured and interpretable way remains an open challenge. In this work, we explore approaches to representing affective behavior in student agents, with the goal of improving the realism and pedagogical value of AI-driven classroom simulations. The implementation is publicly available in a GitHub repository[1]. To contextualise this work, it is necessary to first establish the key concepts underlying AI-driven educational simulations and student modeling.

II. Background

A central concept in artificial intelligence is the computer agent, which can be defined as a system that operates autonomously, perceives its environment, persists over time, adapts to change, and pursues goals. When multiple agents interact within a shared environment, the system is referred to as a multi-agent system, enabling dynamic interactions such as cooperation and competition [10].

Advances in large language models (LLMs) have significantly enhanced the capabilities of such agents. LLM-based agents can generate context-aware responses, maintain conversational coherence, and simulate aspects of human-like reasoning. Additionally, recent approaches enable agents to

[1]https://github.com/erjonk22-pixel/vistta1-ai-classroom-simulation

incorporate memory, reflection, and planning, allowing their behavior to evolve over time [11].

A critical component of such simulations is the representation of student behavior. Effective student modeling requires capturing not only cognitive aspects, such as knowledge and misconceptions [12], [13], but also dispositional traits, such as personality, and dynamic affective states [6], [7], [8]. These factors influence how students respond to instruction, interact with peers, and engage with learning tasks. Incorporating these dimensions into agent design is essential for achieving realistic and pedagogically meaningful simulations [5].

Building on these foundational concepts, prior research has explored a range of approaches to AI in education, agent-based systems, and simulation environments.

III. Related Work

A. Artificial Intelligence in Education

Artificial intelligence has been widely explored in educational contexts, particularly in the development of systems aimed at supporting student learning [12]. A review by Zhai et al. [14] organizes AI in education into three distinct dimensions: a development layer, focused on constructing intelligent learning environments through techniques such as classification and recommendation; an extraction layer, which supports interaction through adaptive feedback and human-computer interaction; and an application layer, which incorporates human-centered elements such as affective computing and immersive learning. Across these dimensions, existing systems predominantly emphasize improving student learning and engagement.

Holmes and Tuomi [15] similarly note that current AI applications are largely centered on supporting learners and optimizing instructional content, with comparatively limited focus on assisting teachers or providing training to manage complex classroom interactions.

B. Large Language Models and Multi-Agent Simulation

Recent developments in large language models (LLMs) have significantly expanded the capabilities of AI agents, enabling natural, context-aware communication, making them ideal for interactive simulation environments.

Park et al. [11], for example, build a generative agent simulation, leveraging an architecture of observation, planning, and reflection in conjunction with an LLM. Through this design, agents are able to store experiences, synthesize higher-level insights, and generate contextually appropriate actions, resulting in believable behavior within a simulated environment.

In addition, recent frameworks have explored structured approaches to multi-agent interaction. CAMEL [16] proposes a role-playing paradigm in which agents are assigned complementary roles, enabling autonomous cooperation. Similarly, AutoGen [17] introduces a conversation-centric framework in which multiple agents collaborate through flexible communication patterns to solve complex problems. These approaches demonstrate how multi-agent systems can utilize the complementary strengths of individual agents to improve reasoning, coordination, and task performance.

Together, these developments highlight the increasing feasibility of constructing multi-agent environments in which diverse and context-dependent behaviors can emerge, providing a foundation for more complex and interactive simulation systems.

C. Emotional Modeling in AI Systems

Emotional dynamics are fundamental to human learning and interaction, influencing attention, motivation, and knowledge acquisition [7], [8]. Modeling affective behavior, as a result, has become an important area of research in artificial intelligence.

The field of affective computing focuses on enabling systems to recognize, interpret, and generate emotional responses [9]. In the context of large language models, recent research has demonstrated that AI systems can generate responses that convey or evoke emotional states during interaction [18]. These capabilities allow agents to produce more expressive, context-sensitive behavior, contributing to more natural and engaging conversations.

D. Simulation-Based Teacher Training

Simulation-based approaches are a proposed method of providing controlled and replicable environments suitable for teacher training. Systems such as EduVerse [5] explore the use of human-in-the-loop, multi-agent environments to simulate classroom scenarios. In these systems, multiple agents interact within a shared environment to model classroom dynamics, enabling users to engage in realistic teaching situations. Such approaches demonstrate the potential of agent-based simulations to support experiential learning by allowing teachers to practice instructional strategies and classroom management in a low-stakes setting.

E. Research Gap

Existing research has demonstrated the potential of AI in education, large language model-based agents, affective computing, and simulation-based teacher training. However, these areas are often explored separately. Educational AI systems primarily focus on supporting student learning, while multi-agent frameworks emphasize general interaction and task completion. Similarly, affective modeling has advanced emotional expression in AI systems but is rarely integrated with structured representations of student knowledge and personality within classroom simulations.

As a result, limited work has explored how knowledge, personality, and affective state can be combined within a unified student model to support realistic classroom behavior. This study addresses that gap by proposing a structured student persona framework that integrates these dimensions within an AI-driven classroom simulation for teacher training.

IV. Research Question

This study investigates whether structured student profiles can produce believable student behavior in an AI-driven classroom

simulation. The main research question is:

RQ1: To what extent can weighted knowledge and personality variables produce believable and distinguishable student personas in an AI classroom simulation?

Affective state variables are included in the system model to support short-term behavioral variation. However, the evaluation primarily focuses on whether knowledge and personality variables contribute to behavior that is perceived as consistent with the intended student personas.

V. Methods

This section describes the classroom simulation and evaluation procedure used to assess whether weighted student profiles can produce believable student behavior. The system architecture is illustrated in Fig. 1. The framework combines structured student representations, task-aware interaction logic, heuristic behavioral decision rules, and LLM-based response generation.

A. System Overview and Student Representation

The system is implemented as an interactive multi-agent classroom simulation. A teacher provides natural-language input through a web interface, and multiple student agents respond according to their internal profiles and the current classroom context. This design follows prior work on educational and classroom simulations, where multiple role-based agents can create more dynamic interaction than a single-agent setup [5], [19].

Each student agent is represented through three main persona components: knowledge, personality, and affective state. These explicit variables are used to make student behavior more controllable and interpretable than relying only on free-form prompt descriptions. Knowledge modeling draws on research emphasizing learner knowledge, misconceptions, and gaps in understanding [12], [13], [20]. Personality variables are motivated by personality-aware student simulation research showing that traits can influence classroom behavior and communication patterns [21]. Affective state variables are motivated by research showing that emotions such as confusion, engagement, and frustration influence learning interaction [6], [8], [7].

Knowledge is represented as a set of mastered concepts together with associated misconceptions. This follows an overlay-style approach, where each student possesses a subset of concepts from a broader curricular knowledge base. Misconceptions are represented as predefined learner-error patterns that may affect reasoning when relevant to the current task. Personality is represented through four stable traits: participation tendency, emotional stability, task orientation, and receptiveness to guidance. Affective state is represented through six dynamic variables: boredom, engagement, confusion, frustration, delight, and surprise.

Two student agents were designed and evaluated for this study. The personas were manually designed to represent distinct but plausible classroom learners. Sophia was designed as a careful high achiever with relatively high task orientation and receptiveness to guidance, while Liam was designed as a shortcut learner with lower task orientation and participation, fewer mastered concepts, and misconceptions related to comma use and run-on sentences. The evaluation focused on whether these intended profile differences were visible to evaluators during interaction.

B. Interaction and Response Pipeline

Each classroom turn follows a structured pipeline. Teacher input is first classified into instructional features such as move type, target, topic, difficulty, and whether a response is expected. The supported move types are question, greeting, praise, correction, information, and acknowledgement. Classification combines LLM-based categorization with rule-based refinement to improve consistency for student-name targeting, group-directed prompts, continuation prompts, and prompts asking multiple students to respond.

The system also performs task detection through rule-based pattern matching. Detected tasks include vocabulary meaning, theme or main-idea questions, sentence construction, spelling, past tense, category examples, and difficult knowledge questions. The active task stores information such as task type, target word or concept, answer rule, attempt count, and most recent target student. This allows the system to interpret follow-up turns such as "try again", "same question", or "now you try" without requiring the teacher to restate the full task.

For each turn, the system computes a student-specific turn context. This context estimates whether the student is likely to answer correctly, partially, incorrectly, or with uncertainty. The decision is task-aware: the system identifies a relevant curricular concept and compares it with the student's mastered concepts. The decision is also adjusted according to task difficulty, confusion, repeated-task pressure, and relevant misconceptions.

The system also estimates response willingness separately from answer correctness. This allows a student to know an answer but still hesitate, or to attempt an answer despite weaker knowledge. Response priority is then determined from a combination of personality traits, current affective state, task context, teacher targeting, and interaction memory. A directly addressed student receives higher priority, while non-targeted students are less likely to answer directed questions. For class-directed prompts, responders are selected from the highest-priority candidates using probabilistic sampling.

These behavioral decision rules are heuristic rather than empirically derived. They were designed to make the influence of knowledge, personality, affective state, and classroom context explicit and adjustable. Therefore, the evaluation does not validate the individual weights or thresholds as psychological measurements. Instead, it evaluates whether the combined model produces behavior that is perceived as believable and consistent with the intended personas.

C. Language Model Use

The system uses an LLM for teacher-input classification and student-response generation. Early prototyping used a local

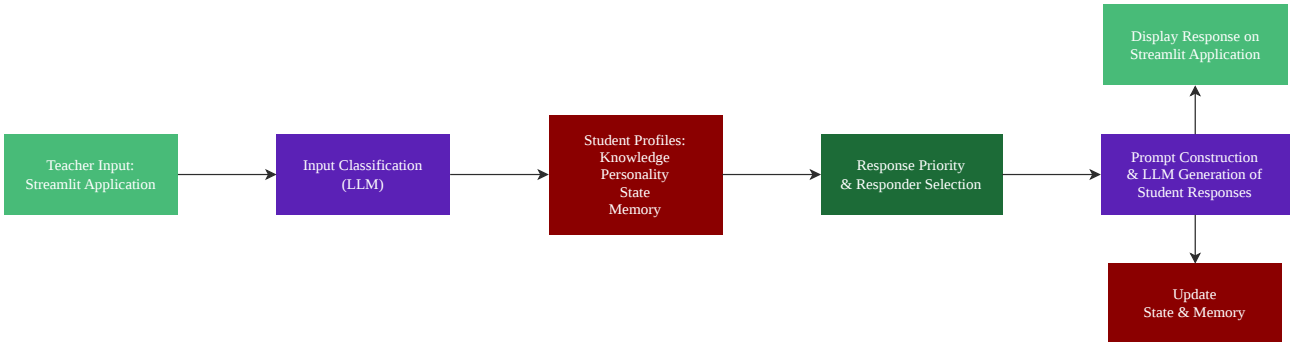


Figure 1. High-level architecture of the classroom simulation system. Teacher input is classified and combined with student representations to model classroom participation, generate student responses, and maintain interaction continuity through state and memory updates.

Llama 3 model through Ollama to support offline experimentation. For the final evaluation, the system used Gemini 2.5 Flash-Lite through the Gemini API. This choice was practical: the simulation required low-latency responses and a cost-efficient model suitable for repeated interactive generation. The model choice should therefore be understood as an engineering decision rather than a claim that Gemini is uniquely optimal for student simulation.

For greetings, praise, corrections, and non-English input, the system uses short predefined reactions to avoid unnecessarily long responses. For question-based turns, a prompt is constructed from the student profile, current turn context, recent classroom context, active task, response style, and answer-quality label. The prompt instructs the model to respond as a Swedish child aged 10–12 learning English using short and simple English. The LLM therefore acts primarily as a language realization component: the structured student model decides the intended response type, while the LLM phrases that decision naturally. Additional prompt rules discourage expert-like answers, rare vocabulary, dictionary-style explanations, and copying from other students.

D. State, Memory, and Evaluation Control

During interaction, the system maintains short-term memory variables such as turns since speaking, silence streak, recent praise, recent correction, and last outcome. These variables are not part of the fixed persona design, but support conversational continuity and later participation decisions.

When uncontrolled, affective state may also change. For example, success may increase delight and engagement, while struggle may increase confusion and frustration. Praise and correction may also influence state depending on the student's receptiveness and emotional stability. These update rules are exploratory and intended to provide short-term behavioral variation rather than validated affective modeling.

Although affective state is included in the system model, the evaluation primarily focused on knowledge and personality. Affective-state recognition was retained as a secondary diagnostic measure to examine whether internal state variables were visible to evaluators. This was included because the system models affective state, but the main research question concerns whether weighted knowledge and personality variables produce believable and distinguishable student personas.

E. Evaluation Design and Data Analysis

The evaluation assessed whether generated student behavior was perceived as believable and consistent with the intended personas. A total of 4 evaluators participated. Evaluators interacted with the simulation under predefined instructions and then completed a questionnaire about the student agents.

The evaluation focused on observable persona consistency rather than the correctness of individual internal weights. Evaluators judged whether students differed in knowledge level, participation tendency, emotional stability, task orientation, receptiveness to feedback, and response behavior. The questionnaire used multiple-select questions and Likert-scale items [22] together with open-ended questions about realism, consistency, and usefulness for teacher training.

Quantitative responses were analyzed by comparing evaluator judgments with the intended persona profiles. Personality-related judgments were mapped to the same $[0, 1]$ scale as the internal profile values. For each personality attribute, alignment was calculated from the absolute difference between the intended value y_i and the evaluator-estimated value $\hat{y}_i$:

$$\text{Alignment}_i = 1 - |\hat{y}_i - y_i|. \tag{1}$$

Overall personality alignment was calculated as the mean alignment across evaluated personality attributes:

$$\text{Overall Alignment} = 1 - \frac{1}{N}\sum_{i=1}^{N} |\hat{y}_i - y_i|, \tag{2}$$

where N is the number of evaluated personality attributes. Higher values indicate stronger alignment between intended and perceived persona traits.

Knowledge and misconception recognition were analyzed by comparing evaluator selections with the intended correct concepts and misconceptions. Recognition accuracy measured the proportion of intended items identified by evaluators, while detection specificity measured the proportion of selected items

that matched the intended profile. Affective state recognition was analyzed similarly, but was interpreted as a secondary diagnostic measure rather than as the main evaluation outcome.

Open-ended responses were analyzed using exploratory thematic coding. Comments were grouped into themes related to realism, age appropriateness, repetition, maturity of responses, distinguishability between students, and usefulness for teacher training. The qualitative analysis was used to contextualize the quantitative results and identify limitations not captured by numeric alignment scores alone.

VI. Results

This section presents the results of the evaluation of the AI student simulation system. Quantitative results are based on responses from four teacher evaluators. Quantitative findings are reported for knowledge and misconception recognition, personality alignment, and affective state recognition. Qualitative findings are also presented through thematic analysis of evaluator feedback from three of the four teacher evaluators.

A. Knowledge and Misconception Recognition

Table I presents evaluator accuracy and specificity scores for the observable knowledge and misconceptions of each student agent.

Table I
Comparison of knowledge and misconception recognition accuracy and detection specificity values between student agents. OKRA: Observable Knowledge Recognition Accuracy; OKDS: Observable Knowledge Detection Specificity; OMRA: Observable Misconception Recognition Accuracy; OMDS: Observable Misconception Detection Specificity.

Student	OKRA	OKDS	OMRA	OMDS
Sophia	0.625	0.313	0.250	0.250
Liam	0.500	0.175	0.000	0.000

Knowledge accuracy scores were somewhat high across evaluated student agents, exceeding knowledge detection specificity in both cases. Evaluators often selected correct concepts for student agents that were not present in their internal representations, as indicated by the lower specificity values.

Furthermore, misconceptions were more difficult for evaluators to recognize than correct concepts, as corresponds to their lower accuracy and specificity values. Misconception recognition scores were lower than knowledge recognition scores for both student agents.

All calculated values were higher for Sophia than for Liam.

B. Personality Alignment

Table II presents evaluator alignment scores for each evaluated student agent. Fig. 2 displays their personality alignment scores for visual comparison.

Personality alignment scores were relatively high across evaluated student agents, with emotional stability and receptiveness to guidance producing more consistently high values than participation tendency and task orientation. The alignment values for the Sophia student agent consistently scored higher than those for the Liam student agent, with the exception of emotional stability.

Table II
Comparison of personality trait alignment scores and overall alignment between student agents. PT: Participation Tendency; ES: Emotional Stability; TO: Task Orientation; RTG: Receptiveness to Guidance; OA: Overall Alignment.

Student	PT	ES	TO	RTG	OA
Sophia	0.975	0.913	0.988	0.900	0.944
Liam	0.700	0.975	0.775	0.863	0.828

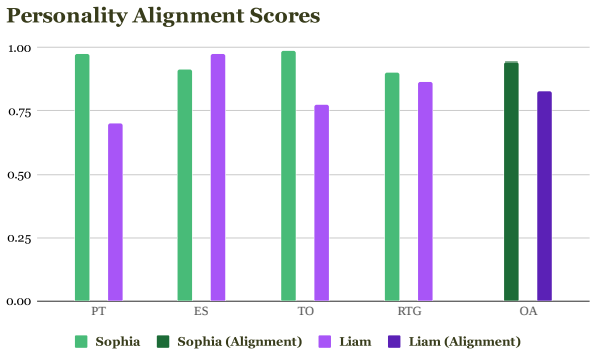


Figure 2. Personality trait alignment scores and overall alignment for each student agent. PT: Participation Tendency; ES: Emotional Stability; TO: Task Orientation; RTG: Receptiveness to Guidance; OA: Overall Alignment.

C. State Recognition

Table III presents evaluator scores for state recognition accuracy and specificity.

Table III
Comparison of state recognition accuracy and detection specificity between student agents. SRA: State Recognition Accuracy; SDS: State Detection Specificity.

Student	SRA	SDS
Sophia	0.250	0.500
Liam	0.250	0.500

The student agents produced identical state recognition scores, which were low across evaluated student agents. Detection specificity scores were higher than recognition accuracy scores for both student agents, meaning that a higher percentage of their selected states were the intended internal representation. Evaluators had difficulty detecting some dominant states, leading to lower accuracy scores.

D. Qualitative Results

Qualitative responses were collected from three teacher evaluators. Responses were analyzed using exploratory thematic coding. Several preliminary themes emerged from evaluator feedback.

Evaluators perceived student behaviors as generally authentic, age-appropriate, and representative of real learners in a classroom. One evaluator responded that student agents behaved in a way they found to be *"very realistic."*

Some particularly thorough responses, however, reduced realism of student agent behavior. For example, one evaluator commented that some responses from student agents seemed *"too mature for their age."*

Further, evaluators highlighted participation patterns such as hesitancy, uncertainty, simple vocabulary, requests for validation from the teacher, and contribution through repetition as realistic. One evaluator described student agents giving the same answer as their peers as authentic to real learners, who often *"don't have an answer, but they want to contribute."*

Some repetitive responses, however, reduced the believability of the system. For instance, one evaluator wrote that some of the responses received from student agents *"were repeated word for word."*

Evaluators identified many potential applications of this system for teacher preparation and classroom practice, especially for asking comprehension questions during classroom discussion. One evaluator found the simulation to be potentially *"very useful in training new teachers and preparing them for real classroom interaction."*

VII. Discussion

The evaluation results indicate that structured student profiles can produce distinguishable behavioral patterns, particularly through personality-related variables. Evaluators were generally able to infer relative differences between student agents, suggesting that stable behavioral traits such as participation tendency and task orientation are more readily perceived in interaction compared to internal cognitive or affective states.

In contrast, knowledge and misconception interpretation was less consistent. Evaluators tended to infer correct knowledge from fluent responses, even when it was not explicitly represented in the internal model. Misconceptions were particularly difficult to detect, especially when they were not observable through spoken interaction, which highlights limitations of the current communication modality and representation design.

Affective state variables were not reliably identified by evaluators, suggesting that emotional dynamics were not sufficiently salient in the current system implementation. This indicates a potential gap between internal state modeling and externally observable behavior in the simulation.

Qualitative feedback further highlighted a trade-off between realism and consistency. While behaviors such as hesitation, simple vocabulary, and uncertainty were perceived as realistic for student agents, issues such as overly mature responses and repeated answers reduced perceived authenticity. These findings suggest that response diversity and age-appropriate language generation are important factors for improving simulation quality.

Overall, the results suggest that structured knowledge and personality variables can support believable student behavior, while affective state modeling and response diversity remain key areas for improvement in future work.

VIII. Conclusion

This study investigated RQ1: to what extent weighted knowledge and personality variables can produce believable student personas in an AI-driven classroom simulation. The results provide preliminary evidence that weighted knowledge and personality variables can contribute to believable student personas in AI-driven classroom simulations, although their effectiveness varies across different dimensions of student behavior. Personality traits were the most consistently recognized dimension, while knowledge recognition was affected by evaluator over-attribution and the limited observability of some misconceptions. Affective states remained largely imperceptible. Despite this, evaluators generally perceived the student agents as authentic and age-appropriate, and unanimously identified potential value for teacher preparation, even though repetitive responses and occasional overly mature language reduced realism in some cases.

Future work should focus on tighter integration of affective and knowledge variables, more precise misconception modeling, and improved response diversity to prevent peer repetition. Reducing LLM influence over parts of the response pipeline may also address issues of overly mature language and behavioral inconsistency. A more diverse set of student personas would better represent the range of learners found in real classrooms. Evaluation could be further strengthened by supplementing questionnaires with interviews or recorded interaction sessions. Despite its limitations, this study provides encouraging evidence that structured student profiles can support realistic AI-driven classroom simulations and contribute to teacher training environments.

References

[1] L. Boström, "What is the problem and how can we solve it? school authorities' perceptions of the shortage of teachers in sweden," *Educational Research for Policy and Practice*, vol. 22, pp. 479–497, 2023.

[2] J. Yin, "Connecting theory and practice in teacher education: English-as-a-foreign-language pre-service teachers' perceptions of practicum experience," *Innovation and Education*, 2019.

[3] F. Amobi and I. Leslie, "Implementing on-campus microteaching to elicit preservice teachers' reflection on teaching actions: Fresh perspective on an established practice," *Journal of the Scholarship of Teaching and Learning*, pp. 27–34, 2009.

[4] S. Ledger and J. Fischetti, "Micro-teaching 2.0: Technology as the classroom," *Australasian Journal of Educational Technology*, 2020.

[5] Y. Ma, S. Hu, Y. Wang, Y. Kang, S. Liu, and K. H. Cheong, "Eduverse: A user-defined multi-agent simulation space for education scenario," *arXiv preprint arXiv:2510.05650*, 2025.

[6] M. H. Immordino-Yang and A. Damasio, "We feel, therefore we learn: The relevance of affective and social neuroscience to education," *Mind, Brain, and Education*, vol. 1, no. 1, pp. 3–10, 2007.

[7] S. D'Mello and A. Graesser, "Dynamics of affective states during complex learning," *Learning and Instruction*, vol. 22, no. 2, pp. 145–157, 2012.

[8] R. Pekrun, "The control-value theory of achievement emotions: Assumptions, corollaries, and implications for educational research and practice," *Educational Psychology Review*, vol. 18, no. 4, pp. 315–341, 2006.

[9] R. W. Picard, *Affective Computing*. MIT Press, 1997.

[10] S. J. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2021.

[11] J. S. Park, J. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023, pp. 1–22.

[12] B. P. Woolf, *Building Intelligent Interactive Tutors: Student-Centered Strategies for Revolutionizing E-Learning*. Morgan Kaufmann, 2009.

[13] J. S. Brown and R. R. Burton, "Diagnostic models for procedural bugs in basic mathematical skills," *Cognitive Science*, vol. 2, no. 2, pp. 155–192, 1978.

[14] X. Zhai *et al.*, "A review of artificial intelligence (ai) in education from 2010 to 2020," *Complexity*, 2021.

[15] W. Holmes and I. Tuomi, "State of the art and practice in ai in education," *European Journal of Education*, vol. 57, no. 4, pp. 542–570, 2022.

[16] G. Li, H. A. A. K. Hammoud, H. Itani, D. Khizbullin, and B. Ghanem, "Camel: Communicative agents for mind exploration of large language model society," *arXiv preprint arXiv:2303.17760*, 2023.

[17] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, and C. Wang, "Autogen: Enabling next-gen llm applications via multi-agent conversations," *arXiv preprint arXiv:2308.08155*, 2024.

[18] Y. Zhang, X. Yang, X. Xu, Z. Gao, Y. Huang, S. Mu, S. Feng, D. Wang, Y. Zhang, K. Song, and G. Yu, "Affective computing in the era of large language models: A survey from the nlp perspective," *Knowledge-Based Systems*, vol. 337, Mar. 2026.

[19] Z. Zhang, D. Zhang-Li, J. Yu, L. Gong, J. Zhou, Z. Hao, J. Jiang, J. Cao, H. Liu, Z. Liu, L. Hou, and J. Li, "Simulating classroom education with llm-empowered agents," *arXiv preprint arXiv:2406.19226*, 2024.

[20] S. Sonkar, X. Chen, N. Liu, R. G. Baraniuk, and M. Sachan, "Llm-based cognitive models of students with misconceptions," *arXiv preprint arXiv:2410.12294*, 2024.

[21] Z. Liu, S. X. Yin, G. Lin, and N. F. Chen, "Personality-aware student simulation for conversational intelligent tutoring systems," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2024, pp. 626–642.

[22] R. Likert, "A technique for the measurement of attitudes," *Archives of Psychology*, vol. 140, pp. 1–55, 1932.